THE EFFECTS OF EXTENDED UNEMPLOYMENT INSURANCE OVER THE BUSINESS CYCLE

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MOTIVATION

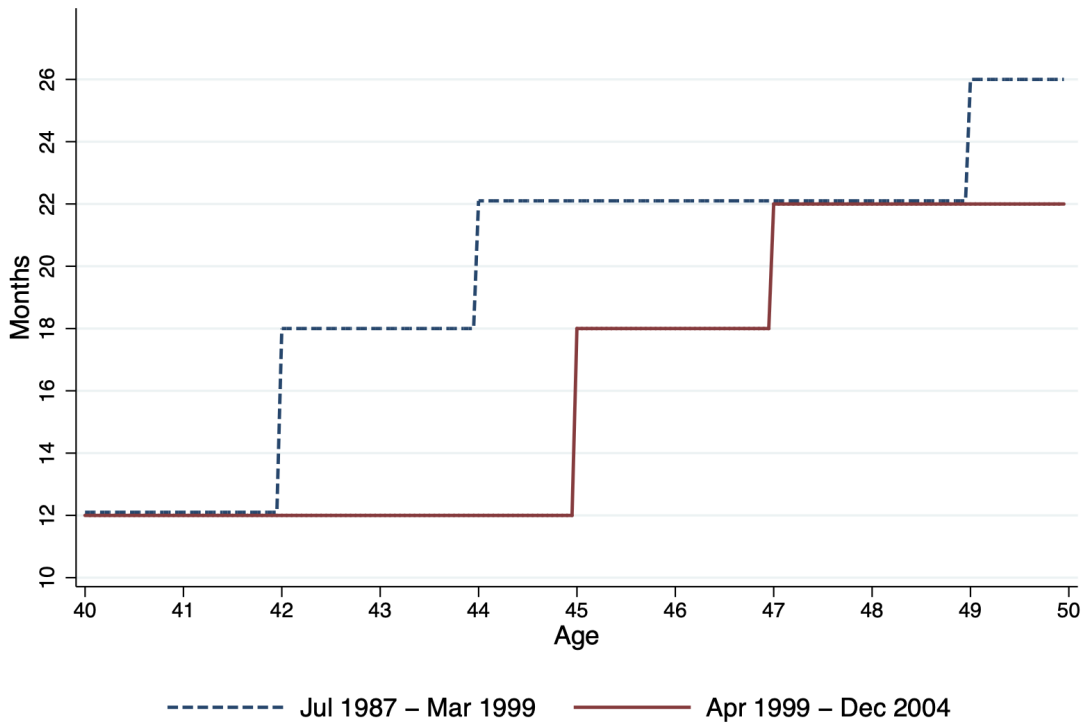
- Recessions trigger statutory extensions of UI benefit duration as spells lengthen
- Common concern: benefit extensions raise nonemployment durations via moral hazard
- Theoretical predictions are contradictory:
  - Larger moral hazard in downturns: skill depreciation compounds when job finding is low (Ljungqvist & Sargent 1998)
  - Smaller moral hazard in downturns: labor rationing makes individual search effort less decisive (Krueger & Meyer 2002)
- Identification challenge: UI extensions are endogenous to labor market conditions

GERMAN UI SYSTEM

- **System basics:** Income replacement for involuntary job losers; 63% of previous net earnings (68 percent with children); eligibility after 12 months of work in the previous 3 years.
- **Policy Lever:** German potential UI duration jumps sharply at exact ages: Pre-1999: 42 (12 to 18 months), 44 (18 to 22), 49 (22 to 26)
- **1999 regime:** From April 1999, a reform lowered durations and shifted thresholds up three years (Eg: 18 months at age 45).
- **Unemployment Assistance:** Unlimited duration; average replacement only about 35% for men, 10% for married women, and 10 to 15% of UI spells transition to it.

FIGURE 1

Figure I: Potential Unemployment Insurance Durations by Period for Workers with High Prior Labor Force Attachment



Notes: The figure shows how potential unemployment insurance (UI) durations vary with age and over time for unemployed individuals workers who had worked for at least 52 months in the last 7 years without intermittent UI spell.

DATA AND SUB-SAMPLE

- **Source**: Universe of German social security records, day-by-day longitudinal data on employment, UI receipt, wages, and benefit levels.
- **Main Sample:** all spells for workers aged 40 to 49 starting July 1987 to December 2004, over 9 million periods; core analysis restricts to high-attachment workers, so potential duration is measured precisely.
- **Representativeness:** the core sample is more male, higher-tenure, higher-earning than the general nonemployed population, but typical of mature recession job losers and comparable to US UI populations of the same ages; results hold in broader samples.

METHODOLOGY

- **Workhorse regression:**

$$Y_{ia} = \beta_0 + \beta_1 D_{(a \geq a^*)} + f(a) + \varepsilon_{it}$$

with $f(a)$ is a linear age control with different slopes at each side of the cutoff. β_1 is the jump in the outcome at the threshold.

- **Design:** Replicate the RD for every year and industry to produce year specific marginal effects, then regress those on cycle measures (GDP growth, unemployment level and change, mass layoffs), with year-level error components.

IDENTIFICATION ASSUMPTIONS

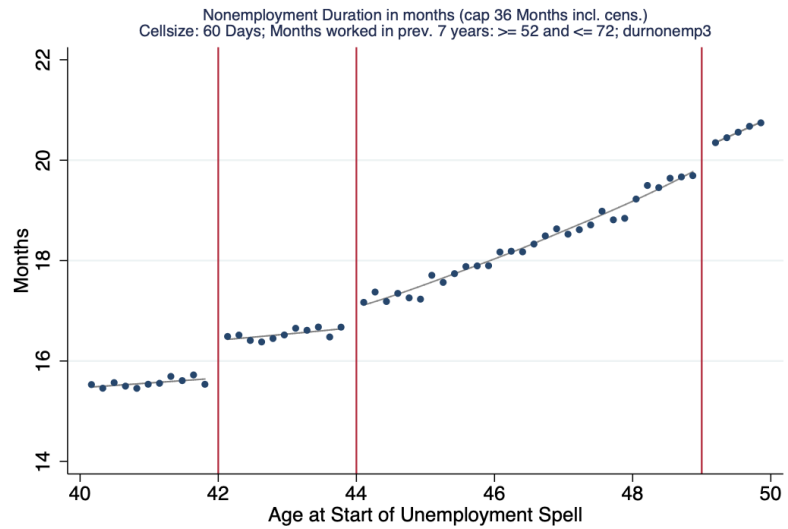
- **Assumption**: everything other than potential UI duration varies smoothly through the age threshold. Workers and firms can influence the timing of job loss and UI claiming, so either could sort around cutoffs
- **Covariate balance**: Re-running the RD with predetermined characteristics as outcomes
- **Density (McCrary test):** a slight excess of spells appears just right of the age 42 and 49 cutoffs (significant but very small)
- **Firm Selection:** the density by job-ending date shows no permanent shift, ruling out systematic layoff of workers with higher entitlements; at most a small local bump at ages 42 and 49

VISUAL CHECK

Figure III: The Effect of Potential Duration in Unemployment Insurance (UI) Benefits on Months of Actual UI Benefit and Months of Nonemployment by Age - Period 1987 to 1999



(a) Actual UI Duration



(b) Nonemployment Duration

RD ESTIMATES

Table II: Regression Discontinuity Estimates of Potential Unemployment Insurance (UI) Benefit Duration (P) on Months of Actual UI Benefit Receipt and Months of Nonemployment

	(1)	(2)	(3)	(4)
	Age bandwidth around age discontinuity			
	2 years	1 year	0.5 years	0.2 years
Panel A: Dependent Variable: Duration of UI Benefit receipt (B)				
D(age>=42)	1.78	1.82	1.73	1.65
	[0.036]**	[0.052]**	[0.072]**	[0.11]**
Effect of 1 add. Month of Benefits $\frac{dB}{dP}$	**0.30**	**0.30**	**0.29**	**0.28**
Observations	452749	225774	112436	45301
D(age>=44)	1.04	1.16	1.13	1.24
	[0.047]**	[0.065]**	[0.092]**	[0.15]**
Effect of 1 add. Month of Benefits $\frac{dB}{dP}$	**0.26**	**0.29**	**0.28**	**0.31**
Observations	450280	225134	112597	45258
D(age>=49)	1.40	1.44	1.44	1.72
	[0.074]**	[0.084]**	[0.12]**	[0.18]**
Effect of 1 add. Month of Benefits $\frac{dB}{dP}$	**0.35**	**0.36**	**0.36**	**0.43**
Observations	329680	217942	109238	43812
Panel B: Dependent Variable: Nonemployment Duration (D)				
D(age>=42)	0.78	0.92	1.04	0.79
	[0.086]**	[0.12]**	[0.17]**	[0.27]**
Effect of 1 add. Month of Benefits $\frac{dD}{dP}$	**0.13**	**0.15**	**0.17**	**0.13**
Observations	452749	225774	112436	45301
D(age>=44)	0.41	0.63	0.62	0.78
	[0.089]**	[0.13]**	[0.18]**	[0.30]*
Effect of 1 add. Month of Benefits $\frac{dD}{dP}$	**0.10**	**0.16**	**0.15**	**0.20**
Observations	450280	225134	112597	45258
D(age>=49)	0.43	0.52	0.56	0.79
	[0.11]**	[0.13]**	[0.19]**	[0.29]**
Effect of 1 add. Month of Benefits $\frac{dD}{dP}$	**0.11**	**0.13**	**0.14**	**0.20**
Observations	329680	217942	109238	43812

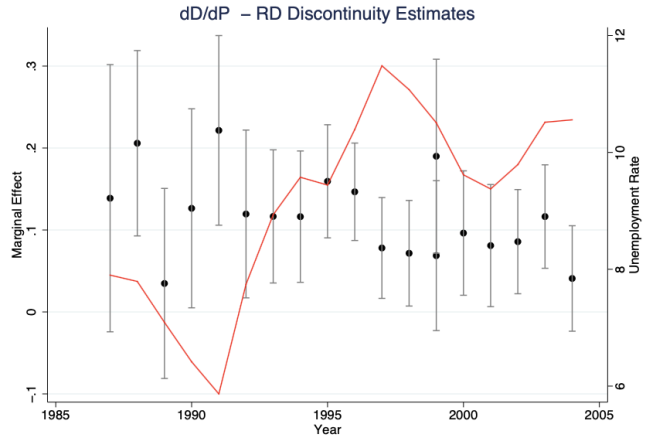
Notes: The coefficients estimate the magnitude of the change in benefit or Nonemployment duration at the age threshold. Each coefficient is estimated in a separate RD regression that controls linearly for age with different slopes on each side of cutoff. Standard errors (in parentheses) are clustered at the day level (* P<.05, ** P<.01).
At the 42-year discontinuity, potential UI benefit durations (P) increase from 12 to 18 months, at the 44-year discontinuity from 18 to 22 months and at the 49-year discontinuity from 22 to 26 months. The sample consists of individuals starting unemployment insurance spells between July 1987 and March 1999, who had worked for at least 52 months in the last 7 years without intermittent UI spells. For the age 49 cutoff and two-year bandwidth column, the regression only includes individuals between 47 and 50 years of age, due to an early retirement discontinuity at age 50.

AVERAGE EFFECT OF UI EXTENSIONS

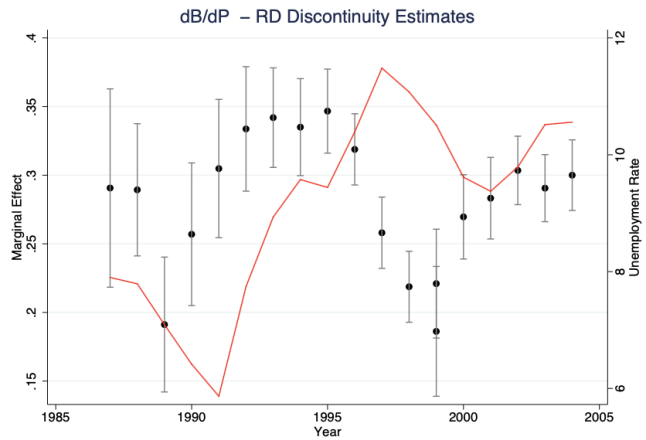
- **Average effect:** Per extra month of potential benefits, workers stay on UI about 9 to 12 days longer, but stay out of work only 3 days longer.
- The gap is explained by **exhaustion:** about 28% of workers just left of the age 42 cutoff exhaust their 12 months of benefits, and among exhaustees only 8% actually return to employment.
- For a large group of workers, an extension of potential duration is purely extra coverage: they were going to stay unemployed past the old cutoff regardless, and the extension simply keeps paying them. That is why the coverage effect (9 to 12 days) exceeds the behavioral effect (3 days). This distinction is the foundation of the welfare analysis later.

RD ESTIMATES

Figure VI: Variation in Regression Discontinuity Estimates of Marginal Effects of Potential Unemployment Insurance Duration at the Age 42 and Age 45 Discontinuities over Time



(a) Effect of Pot. UI Durations on Nonemployment Durations $\frac{dD}{dP}$ and the Unemployment Rate



(b) Effect of Pot. UI Durations on Actual UI Durations $\frac{dB}{dP}$ and the Unemployment Rate

Notes: Each dot in the bottom figure corresponds to a rescaled marginal effect of one month additional potential UI duration estimated at an age cutoff in one year between 1987 and 2004 at the age 42 (before the 1999 reform) and the age 45 (after the 1999 reform) cutoff, where pot. UI durations increased from 12 to 18 months. Since the 1999 reform occurred during the year there are 2 estimates for 1999. The samples are described in Figures II and IV. The line shows the German unemployment rate in each year.

RESULTS OVER THE CYCLE

- The nonemployment effect (dD/dP) remains stable across cyclical conditions: the labor-supply effect of UI extensions does not rise in recessions; if anything it weakly declines (correlations with GDP growth and unemployment changes point the same way).
- **Coverage rises in recessions:** the effect of extensions on actual benefit duration (dB/dP), driven by the exhaustion rate, increases significantly in downturns; a 2σ rise in ΔU raises the marginal effect by 25 to 30%.
- Countercyclical moral hazard ratio:

$$\text{Effective Moral Hazard} = \frac{dD/dP}{dB/dP} \quad \text{falls significantly in downturns}$$

ROBUSTNESS CHECKS

- **Results robust to:** smaller bandwidths, lower-bound reassignment of manipulated observations, restricting to UI claims within 15 days of job loss, reweighting to hold claimant characteristics constant, and samples without experience restrictions
- **Interaction with Unemployment Assistance (UA):** Claimants with higher predicted UA receipt exhibit modestly higher durations
- Married women have near-zero UA eligibility due to household means-testing: Estimated responses for married women are close to men's estimates
- Confirms that post-exhaustion UA receipt does not drive cyclical findings

CONCLUSION AND POLICY IMPLICATIONS

- Core empirical findings across 20 years:
  - Labor supply response ($dD/dP \approx 0.10$) is modest and cyclically stable
  - Marginal coverage (dB/dP) increases markedly during recessions
  - Effective moral hazard ratio (dD/dB) is countercyclical
- Welfare and policy implications:
  - The economic case for UI extensions in downturns is stronger than cycle-average estimates suggest
  - In downturns, extensions provide greater insurance coverage per month of behavioral distortion